

TABLA 1.2:

Characteristic Value

Number of sockets per node 2

Number of cores per socket 16

Number of threads per core 2

Maximum core frequency 3.4 GHz (CPU max MHz: 3400.0000)

L1-I cache size (per-core) 32 KB

L1-D cache size (per-core) 48 KB

L2 cache size (per-core) 1.25 MB ($\approx$ 1280 KB)

Last-level cache size (per-socket) 24 MB

Main memory size (per socket) $\approx$ 63 GB (NUMA node0: 63 GB, node1: 63 GB)Main memory size (per node) $\approx$ 126 GB (suma de los dos sockets)

Cache line size (in bytes) 64 B (valor típico de Intel; también aparece en cache_line_size-?.txt)

Characteristic	Value
Number of sockets per node	2
Number of cores per socket	16
Number of threads per core	2
Maximum core frequency	3.4 GHz (CPU max MHz: 3400.0000)
L1-I cache size (per-core)	32 KB
L1-D cache size (per-core)	48 KB
L2 cache size (per-core)	1.25 MB ($\approx$ 1280 KB)
Last-level cache size (per-socket)	24 MB
Main memory size (per socket)	$\approx$ 63 GB (NUMA node0: 63 GB, node1: 63 GB)
Main memory size (per node)	$\approx$ 126 GB (suma de los dos sockets)
Cache line size (in bytes)	64 B (valor típico de Intel; también aparece en cache_line_size-?.txt)

1.5

Interactively execute the OpenMP code with 1, 2, 4, 8, 16, and 32 threads using 109 iterations:

C/C++

```
paco1114@boada-12:~/lab1/pi$ ./run-seq.sh pi_seq 1000000000
icx -g -O3 -std=c99 -march=native pi_seq.c -o pi_seq
Number pi after 1000000000 iterations = 3.141592653589768
Execution time (secs.): 0.696293
0.69user 0.00system 0:00.70elapsed 99%CPU (0avgtext+0avgdata
5120maxresident)k
40inputs+0outputs (0major+408minor)pagefaults 0swaps
```

Submit the execution script to the queuing system via:
sbatch submit-omp.sh <name> <iterations> <threads>

C/C++

```
paco1114@boada-12:~/lab1/pi$ sbatch submit-seq.sh pi_seq 1000000000  
Submitted batch job 312147
```

PUNTO 1.5

Interactive timing information

```
paco1114@boada-11:~/lab1/pi$ make pi_omp  
icx -g -O3 -std=c99 -march=native -qopenmp pi_omp.c -o pi_omp  
paco1114@boada-11:~/lab1/pi$ ./run-omp.sh pi_omp 1000000000 1  
icx -g -O3 -std=c99 -march=native -qopenmp pi_omp.c -o pi_omp  
Number pi after 1000000000 iterations = 3.141592653589768  
Execution time (secs.): 0.687332  
0.69user 0.00system 0:00.69elapsed 99%CPU (0avgtext+0avgdata 7360maxresident)k  
40inputs+0outputs (0major+606minor)pagefaults 0swaps  
paco1114@boada-11:~/lab1/pi$ ./run-omp.sh pi_omp 1000000000 2  
icx -g -O3 -std=c99 -march=native -qopenmp pi_omp.c -o pi_omp  
Number pi after 1000000000 iterations = 3.141592653589738  
Execution time (secs.): 0.688812  
1.37user 0.00system 0:00.70elapsed 197%CPU (0avgtext+0avgdata 7360maxresident)k  
40inputs+0outputs (0major+619minor)pagefaults 0swaps  
paco1114@boada-11:~/lab1/pi$ ./run-omp.sh pi_omp 1000000000 4  
icx -g -O3 -std=c99 -march=native -qopenmp pi_omp.c -o pi_omp  
Number pi after 1000000000 iterations = 3.141592653589855  
Execution time (secs.): 0.726902  
1.30user 0.15system 0:00.73elapsed 197%CPU (0avgtext+0avgdata 7360maxresident)k  
40inputs+0outputs (0major+631minor)pagefaults 0swaps  
paco1114@boada-11:~/lab1/pi$ ./run-omp.sh pi_omp 1000000000 8  
icx -g -O3 -std=c99 -march=native -qopenmp pi_omp.c -o pi_omp  
Number pi after 1000000000 iterations = 3.141592653589883  
Execution time (secs.): 0.708963  
1.33user 0.09system 0:00.72elapsed 197%CPU (0avgtext+0avgdata 7360maxresident)k  
40inputs+0outputs (0major+664minor)pagefaults 0swaps  
paco1114@boada-11:~/lab1/pi$ ./run-omp.sh pi_omp 1000000000 16  
icx -g -O3 -std=c99 -march=native -qopenmp pi_omp.c -o pi_omp  
Number pi after 1000000000 iterations = 3.141592653589800  
Execution time (secs.): 0.700319  
1.37user 0.03system 0:00.71elapsed 197%CPU (0avgtext+0avgdata 7680maxresident)k  
40inputs+0outputs (0major+725minor)pagefaults 0swaps  
paco1114@boada-11:~/lab1/pi$ ./run-omp.sh pi_omp 1000000000 32  
icx -g -O3 -std=c99 -march=native -qopenmp pi_omp.c -o pi_omp  
Number pi after 1000000000 iterations = 3.141592653589813  
Execution time (secs.): 0.710430  
1.34user 0.08system 0:00.72elapsed 197%CPU (0avgtext+0avgdata 8000maxresident)k  
40inputs+8outputs (0major+833minor)pagefaults 0swaps
```

Queued timing information

```
paco1114@boada-11:~/lab1/pi$ cat time-pi_omp-1-boada-18  
0.69user 0.00system 0:00.72elapsed 97%CPU (0avgtext+0avgdata 7168maxresident)k  
40inputs+8outputs (0major+614minor)pagefaults 0swaps  
paco1114@boada-11:~/lab1/pi$ cat time-pi_omp-2-boada-18  
0.70user 0.00system 0:00.37elapsed 188%CPU (0avgtext+0avgdata 7168maxresident)k  
40inputs+8outputs (0major+626minor)pagefaults 0swaps  
paco1114@boada-11:~/lab1/pi$ cat time-pi_omp-4-boada-18
```

```

0.69user 0.01system 0:00.22elapsed 318%CPU (0avgtext+0avgdata 7168maxresident)k
40inputs+8outputs (0major+645minor)pagefaults 0swaps
paco1114@boada-11:~/lab1/pi$ cat time-pi_omp-8-boada-18
0.69user 0.00system 0:00.11elapsed 609%CPU (0avgtext+0avgdata 7168maxresident)k
40inputs+8outputs (0major+681minor)pagefaults 0swaps
paco1114@boada-11:~/lab1/pi$ cat time-pi_omp-16-boada-18
0.70user 0.01system 0:00.07elapsed 988%CPU (0avgtext+0avgdata 7168maxresident)k
40inputs+8outputs (0major+757minor)pagefaults 0swaps
paco1114@boada-11:~/lab1/pi$ cat time-pi_omp-32-boada-18
0.70user 0.01system 0:00.05elapsed 1405%CPU (0avgtext+0avgdata 7680maxresident)k
40inputs+16outputs (0major+898minor)pagefaults 0swaps

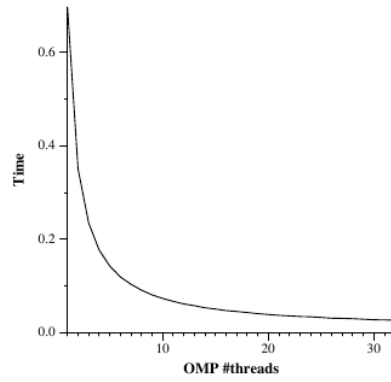
```

TABLA 1.3

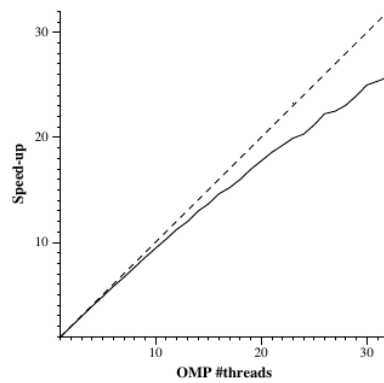
Threads	Interactive	Queued
1	0.69 / 0.00 / 0.69 / 99%	0.69 / 0.00 / 0.72 / 97%
2	1.37 / 0.00 / 0.70 / 197%	0.70 / 0.00 / 0.37 / 188%
4	1.30 / 0.15 / 0.73 / 197%	0.69 / 0.01 / 0.22 / 318%
8	1.33 / 0.09 / 0.72 / 197%	0.69 / 0.00 / 0.11 / 609%
16	1.37 / 0.03 / 0.71 / 197%	0.70 / 0.01 / 0.07 / 988%
32	1.34 / 0.08 / 0.72 / 197%	0.70 / 0.01 / 0.05 / 1405%

1.6

Imagen de strong scability

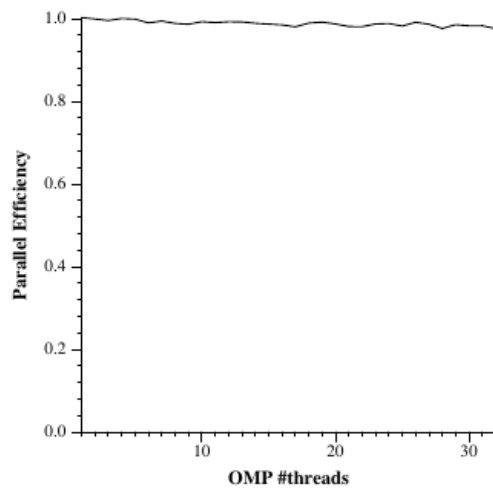


paco1114
Min elapsed execution time
Wed Sep 17 09:34:50 AM CEST 2025



paco1114
Speed-up wrt sequential time
Wed Sep 17 09:34:50 AM CEST 2025

Imagen de weak scability



paco1114
Parallel Efficiency w.r.t. one thread (weak scaling)
Wed Sep 17 09:39:26 AM CEST 2025